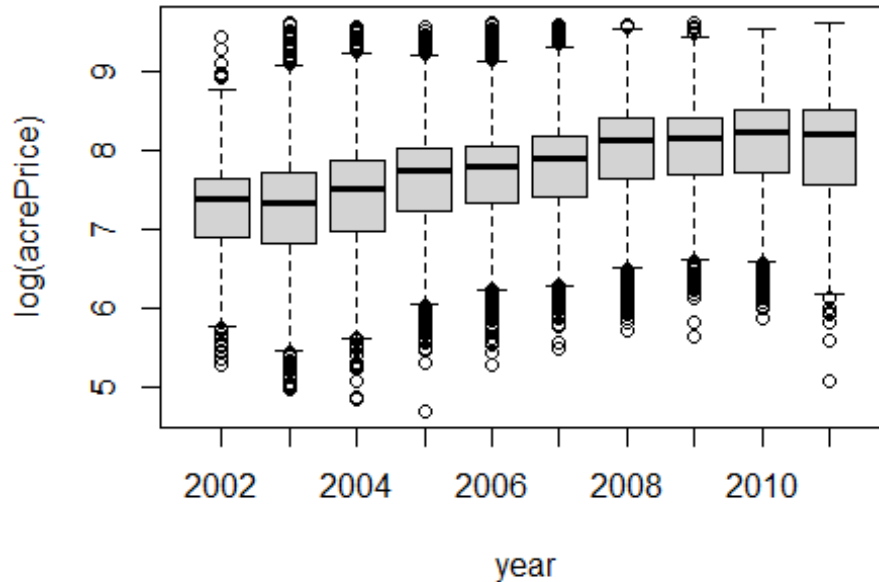


Assignment-7.R

ganga

2022-11-08

```
#problem-5.4  
data(MinnLand)  
# 5.4.1  
boxplot(log(acrePrice)~year,MinnLand)
```



```
# housing sales prices in the United States were generally increasing from  
# about 2002-2006 and began to fall in the beginning of 2007 but  
# its not the same case in our pattern, prices were increasing after 2007  
# in minnland farm sales because the median of prices were increasing.
```

```
#5.4.2  
m1<-as.factor(MinnLand$year)  
m2<-lm(log(acrePrice)~m1,MinnLand)  
summary(m2)  
  
##  
## Call:  
## lm(formula = log(acrePrice) ~ m1, data = MinnLand)  
##
```

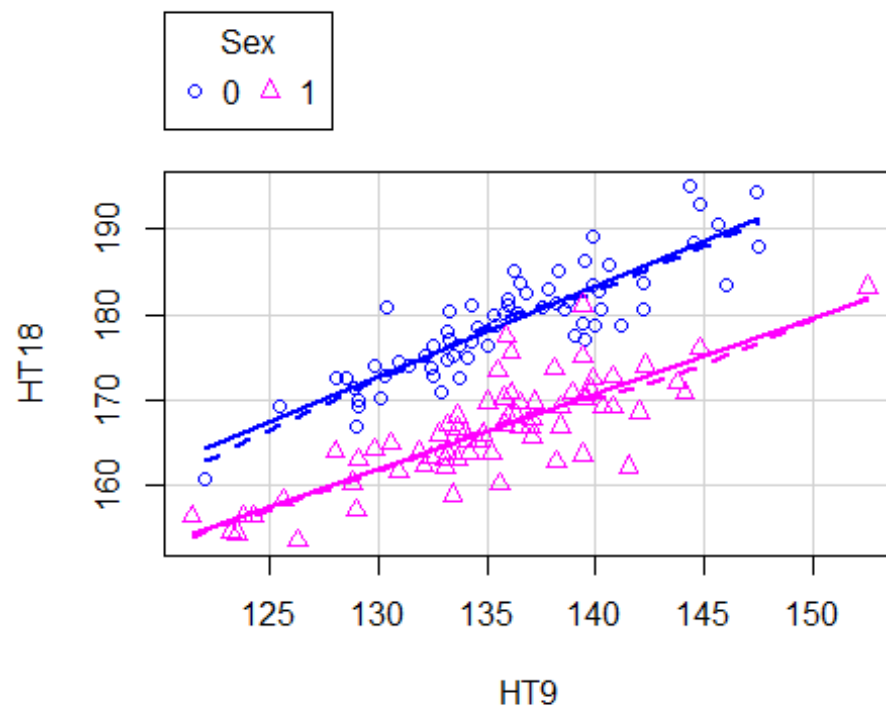
```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.9499 -0.3785  0.1301  0.4354  2.3456
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   7.27175     0.02848 255.345 < 2e-16 ***
## m12003        -0.00155     0.03207  -0.048  0.961
## m12004         0.14794     0.03155   4.689 2.76e-06 ***
## m12005         0.36026     0.03176  11.343 < 2e-16 ***
## m12006         0.39392     0.03195  12.329 < 2e-16 ***
## m12007         0.47682     0.03186  14.965 < 2e-16 ***
## m12008         0.68364     0.03162  21.620 < 2e-16 ***
## m12009         0.71407     0.03355  21.284 < 2e-16 ***
## m12010         0.75733     0.03260  23.231 < 2e-16 ***
## m12011         0.72071     0.03526  20.437 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6775 on 18690 degrees of freedom
## Multiple R-squared:  0.1293, Adjusted R-squared:  0.1289
## F-statistic: 308.5 on 9 and 18690 DF,  p-value: < 2.2e-16

# The intercept is positive and the value of the intercept is 7.27175
# the other coefficients shows the value with respect to the predictor(year)
# to the log(acreprice) with considering the other years.
qt(1-0.05/2,18690)

## [1] 1.960091

# by using t value and quartile, for the year 2003 we donot reject the
# null hypothesis and its not significance and remaining all years we reject
# the null hypothesis and all significant.

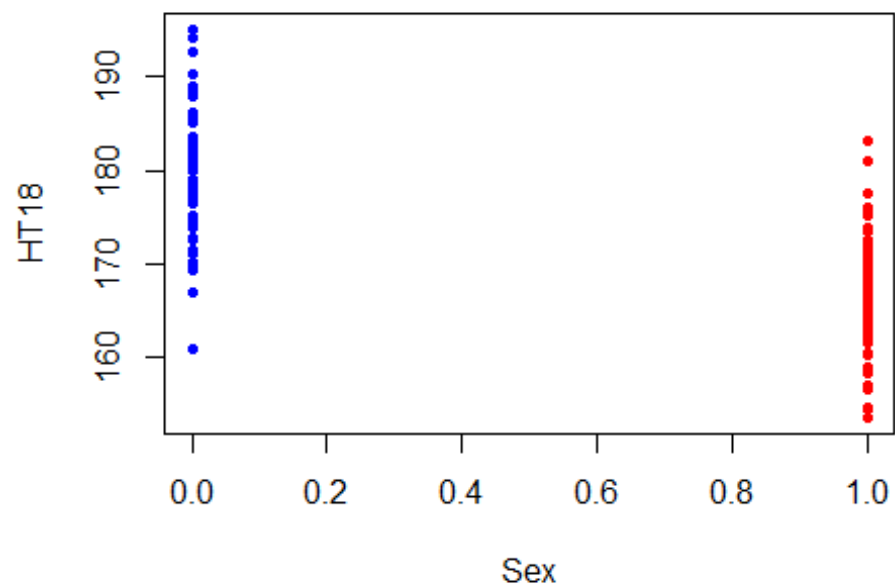
# problem-5.14
#5.14.1
scatterplot(HT18~HT9|Sex,data = BGSall)
```



*# Given scatter plot shows the information of better fit regression lines for
males and females the effect is same in both the lines it looks like
parallel.*

#5.14.2

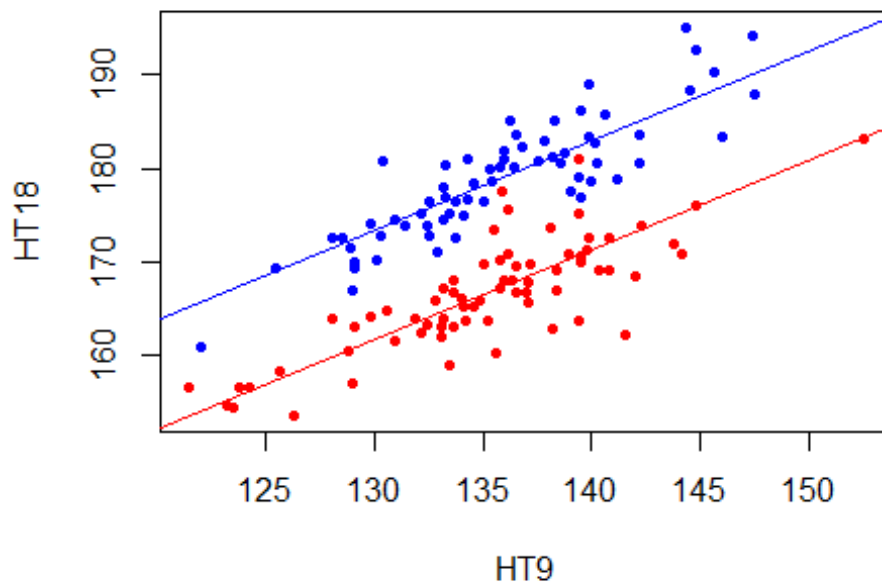
```
with(BGSall, plot(HT18~Sex+HT9, col=ifelse(Sex=="0", "blue", "red"), pch=20))
```



```
m2 <- with(BGSall, lm(HT18~Sex+HT9))
m2$coef

## (Intercept)      Sex      HT9
## 48.5173098 -11.6958406  0.9600564

abline(m2$coef[1],m2$coef[3],col="blue")
abline(m2$coef[1]+m2$coef[2],m2$coef[3],col="red")
```



the given parallel regression model describes the relation between HT18
 # and sex using HT9. for the regression the effect is constant(fixed) also
 the slope
 # the difference of height at the age of 18 between males and females at a
 given height(age-9)
 # is 11.6958406. 1 unit increase in HT9 then result in HT18 by 0.9600564 cm.

#5.14.3

summary(m2)

```
##
## Call:
## lm(formula = HT18 ~ Sex + HT9)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.4694  -2.0952  -0.0136   1.7101  10.4467
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  48.51731    7.33385   6.616 8.27e-10 ***
## Sex         -11.69584    0.59036  -19.811 < 2e-16 ***
## HT9          0.96006    0.05388   17.819 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.432 on 133 degrees of freedom
```

```
## Multiple R-squared:  0.8516, Adjusted R-squared:  0.8494
## F-statistic: 381.7 on 2 and 133 DF,  p-value: < 2.2e-16

#obtaining 95% Confidence Interval
# alpha = 0.05
# quantile of the t distribution
qt(1-0.05/2, 133)

## [1] 1.977961

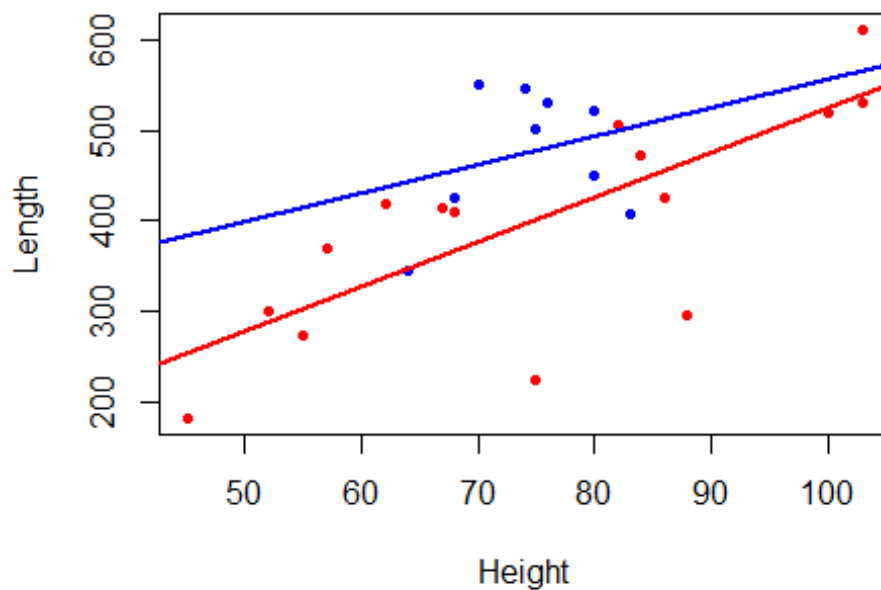
# Upper Limit
-11.69584 + qt(1-0.05/2, 133)* 0.59036

## [1] -10.52813

# Lower Limit
-11.69584 - qt(1-0.05/2, 133)* 0.59036

## [1] -12.86355

#problem 5.16
data(cathedral)
with(cathedral, plot(Length~Height, col=ifelse(Type=="Romanesque", "blue",
"red"), pch=20))
with(cathedral[cathedral$Type=="Gothic",],
abline(lm(Length~Height),col="red",lwd=2))
with(cathedral[cathedral$Type=="Romanesque",],
abline(lm(Length~Height),col="blue",lwd=2))
```



```

m3 <- with(cathedral, lm(Length~Type+Height))
summary(m3)

##
## Call:
## lm(formula = Length ~ Type + Height)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -175.85  -31.51   19.80   55.85   96.96
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    39.584     83.544   0.474 0.640306
## TypeRomanesque  77.268     32.917   2.347 0.028315 *
## Height         4.817       1.074   4.484 0.000185 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 78.95 on 22 degrees of freedom
## Multiple R-squared:  0.5307, Adjusted R-squared:  0.488
## F-statistic: 12.44 on 2 and 22 DF, p-value: 0.0002432

m4 <- with(cathedral, lm(Length~Type+I(Height^2)))
summary(m4)

##
## Call:
## lm(formula = Length ~ Type + I(Height^2))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -165.92  -36.35   20.71   59.72   96.77
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.145e+02  4.751e+01   4.514 0.000172 ***
## TypeRomanesque 8.605e+01  3.340e+01   2.577 0.017211 *
## I(Height^2)    3.137e-02  7.154e-03   4.385 0.000235 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 79.78 on 22 degrees of freedom
## Multiple R-squared:  0.5208, Adjusted R-squared:  0.4772
## F-statistic: 11.95 on 2 and 22 DF, p-value: 0.0003061

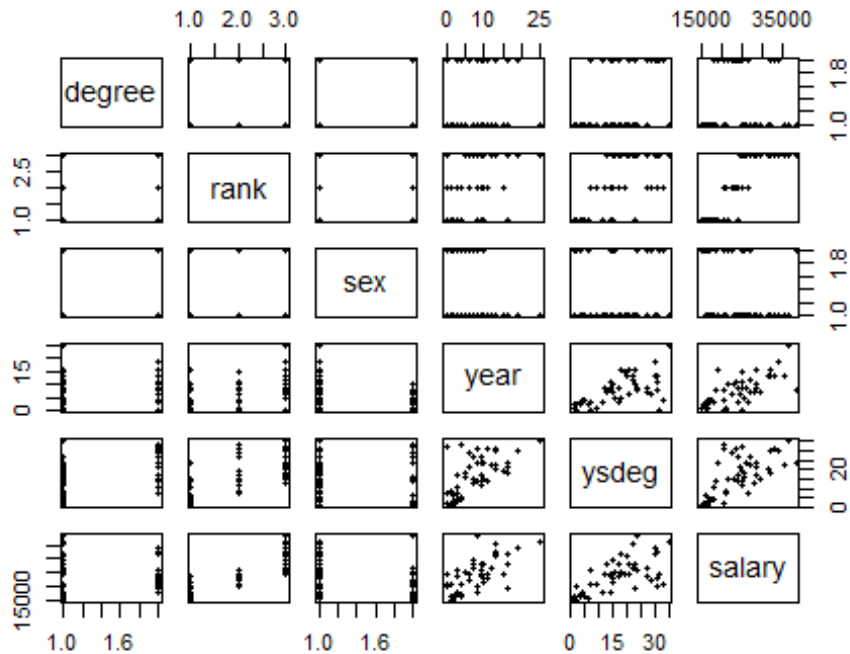
xM <- -(m4$coef[2]/(2*m4$coef[3]))

#PROBLEM 5.17
data(salary)

```

#5.17.1

```
pairs(salary,pch=20)
```



we can see some regression trend on the variables salary,ysdeg and year.

#5.17.2

```
m5<-with(salary,lm(salary~sex))
summary(m5)
```

```
##
## Call:
## lm(formula = salary ~ sex)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8602.8 -4296.6 -100.8  3513.1 16687.9
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    24697         938  26.330  <2e-16 ***
## sexFemale      -3340        1808  -1.847   0.0706 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5782 on 50 degrees of freedom
## Multiple R-squared:  0.0639, Adjusted R-squared:  0.04518
## F-statistic: 3.413 on 1 and 50 DF, p-value: 0.0706
```



```

qt(1-0.05/2,50)

## [1] 2.008559

# the null hypothesis is the mean salary for men and women is the same
# the alternative hypothesis is the mean salary of men and women is not same
and
# we test the hypothesis then the t test result in not rejecting the null
hypothesis.
#5.17.3
#confidence interval
# Upper limit
-3340 + qt(1-0.05/2, 50)* 1808

## [1] 291.4749

# Lower limit
-3340 - qt(1-0.05/2, 50)* 1808

## [1] -6971.475

#5.17.4
m6<-lm(salary~.-rank,data = salary)
summary(m6)

##
## Call:
## lm(formula = salary ~ . - rank, data = salary)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8146.9 -2186.9  -491.5   2279.1 11186.6
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  17183.57    1147.94   14.969  < 2e-16 ***
## degreePhD    -3299.35     1302.52   -2.533  0.014704 *
## sexFemale    -1286.54     1313.09   -0.980  0.332209
## year           351.97      142.48    2.470  0.017185 *
## ysdeg         339.40       80.62    4.210  0.000114 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3744 on 47 degrees of freedom
## Multiple R-squared:  0.6312, Adjusted R-squared:  0.5998
## F-statistic: 20.11 on 4 and 47 DF, p-value: 1.048e-09

```